WATERJET CUTTING

The ultra-high-pressure cold-cutting process — physics, machine anatomy, cutting parameters, history & applications

accuracy ±0.05 mm

max thick 300 mm

87,000 psi 6,000 bar

900 m/s Mach 3

WHAT IS IT?

Waterjet cutting is a **cold machining process**: a jet of water pressurised to **6,000 bar** is forced through a tiny orifice at supersonic speed and erodes material. Adding **garnet** (AWJ) cuts steel, stone and glass to **300 mm** thick — no heat-affected zone, no tool wear, no thermal distortion.

ANATOMY — THE CUTTING HEAD abrasive waterjet cross-section

- Abrasive inlet — garnet feed
- Mixing chamber — venturi vacuum
- Orifice (jewel) — Ø 0.08–0.5 mm
- Catch tank — water + recycled garnet
- Jet ≈ 900 m/s — erosion zone
- Focusing tube — WC, Ø 0.5–1.5 mm

- **Kerf & taper** — slot ≈ tube Ø + 0.2 mm; thick cuts taper; 5-axis tilt (±60°) or compensation fixes it.
- **No melting** — mechanical erosion, not thermal: **zero HAZ**, part stays below ≈ 90 °C.
- **Momentum** — grains strike at shallow angles; micro-chipping + fatigue erode the part, millions/s.
- **Venturi** — the fast jet creates a vacuum that sucks garnet into the stream.
- **Bernoulli** — v = √(2P/ρ): at 4,000 bar the jet exits at ≈ 900 m/s (Mach 3).
- **Energy conversion** — the pump compresses water; energy is stored as pressure.

MACHINE ANATOMY — KEY COMPONENTS

- **CNC gantry** — servo X/Y/Z + tilt head, 20 m/min traverse, ±0.05 mm repeatability.
- **Abrasive feed** — hopper → metering valve → hose, 0.2–1.0 kg/min.
- **Water plant** — filtration + reverse osmosis protects orifice & seals.
- **HP pump** — intensifier (to 6,000 bar) or direct-drive crank (to ≈ 3,500 bar, more efficient).

PURE WATERJET vs ABRASIVE (AWJ)

Feature	Pure waterjet	Abrasive waterjet
Cut medium	Water only	Water + garnet
Materials	Foam, rubber, food, paper, leather, insulation	Steel, Ti, stone, glass, ceramics, composites
Max thickness	≈ 25 mm (soft)	to 300 mm (steel 150–200)
Cut mechanism	Droplet impact erosion	Abrasive particle erosion
Edge finish	Clean, striation-free	Ra 3–12 µm
Running cost	Low — water + electricity	Medium — garnet US$0.3–1.5/kg

KEY CUTTING PARAMETERS

Parameter	Typical	Effect
Water pressure	3,000–6,000 bar	↑ pressure = faster, smoother
Orifice Ø	0.08–0.5 mm	smaller = finer jet, slower
Focusing tube Ø	0.5–1.5 mm	sets kerf width (≈ +0.2 mm)
Abrasive flow	0.2–1.0 kg/min	↑ abrasive = faster (limit)
Traverse speed	0.5–6 m/min	slower = smoother (Ra 3–12 µm)
Standoff distance	1–3 mm	larger = more flare & taper
Garnet mesh	80 std (50–220)	finer mesh = smoother finish
Piercing	pressure ramp + slow feed	clean start hole, no blast-out

HISTORY — PAPER MILLS TO 6,000 BAR

Today 6,000 bar pumps, robotic & micro waterjets (kerf < 0.1 mm), lights-out automation.

1990s 5-axis CNC heads, 4,000 bar intensifiers and CAD/CAM integration arrive.

1983 Dr. Mohamed Hashish invents the abrasive waterjet — metal cutting is unlocked.

1979 Flow Systems installs industrial systems; Boeing adopts waterjet for aerospace.

1971 McCartney Mfg. sells the first commercial waterjet systems (paper & cardboard).

1958 Dr. Norman Franz (U. Michigan) pioneers ultra-high-pressure jets for wood.

1930s–50s Industry uses low-pressure jets for paper cutting and coal mining.

ADVANTAGES

- ✓ Eco: water + garnet (recyclable ×5)
- ✓ Small kerf; stack sheets; pierce anywhere
- ✓ ±0.05 mm accuracy; Ra 3–12 µm edges
- ✓ No tool wear or cutting force
- ✓ Zero HAZ: no distortion, no fire risk
- ✓ Cuts almost any material — to 300 mm

LIMITATIONS

- ✗ No tempered glass / brittle ceramics
- ✗ Taper/striations on thick or fast cuts
- ✗ Loud 85–100 dB — enclosure required
- ✗ Wet: rust, splash, drainage needed
- ✗ High capital cost; garnet consumable
- ✗ Slow for thin metal vs laser / plasma

GARNET — THE ABRASIVE

- **Rate & cost:** 0.2–1.0 kg/min, ≈ US$0.3–1.5/kg; recyclable 3–5×. Alt: olivine / slag.
- **Mesh:** 80 = standard • 120 = finer edge • 50 = fast roughing • 220 = micro-finish.
- **Why garnet?** Almandine — Mohs 7.5–8, sharp angular grains, SG ≈ 4.1; shatters into fresh edges.

WATERJET vs LASER vs PLASMA vs CNC MILL

Criteria	Waterjet	Laser	Plasma	CNC mill
HAZ	**None**	Small	Large	None
Max steel	**300 mm**	20–25 mm	50–80 mm	Tool-dep.
Edge qual.	Good–exc.	Excellent	Fair–rough	Excellent
Kerf width	≈ 1 mm	0.2–0.4 mm	2–5 mm	Cutter Ø
Thin steel	Slow–med	Very fast	Fast	Slow
Materials	Any	Metal/Wood	Conductive	Most
Env.	Wet, noisy	Fumes	Glare	Chips

MATERIALS & APPLICATIONS

- **Foam** — thick, stacked sheets
- **Armour** — no heat-hardened edge
- **Food** — frozen & baked slicing
- **Aerospace** — Ti, Inconel, CFRP
- **Electronics** — PCB depaneling
- **Glass** — signage & lettering
- **Gaskets** — rubber, cork, PTFE
- **Stone** — marble inlay & mosaic

KEY TERMS: HAZ — heat-affected zone • Kerf — width of the cut slot • Taper — kerf narrowing through the thickness • Orifice — jewel nozzle that forms the jet • Focusing tube — carbide tube that aligns the jet • Standoff — nozzle-to-part gap • Traverse speed — table feed rate • Garnet — natural abrasive (almandine) • Intensifier — hydraulic booster pump • Venturi — vacuum that feeds the abrasive • AWJ — abrasive waterjet • 5-axis — tiltable cutting head (±60°)

FORMULAS: Bernoulli v = √(2P/ρ) • Pump power ≈ P × Q (pressure × flow) • Cutting rate rises with power • Abrasive mass flow 0.2–1.0 kg/min • Jet exit ≈ Mach 3 at 4,000 bar • Standoff 1–3 mm • Kerf ≈ tube Ø + 0.2 mm • Taper ≈ 0.5–1° per 25 mm • Water is incompressible — the jet stores ~1 GJ/m³ of energy.